EX.NO:8(a) TEST THE SOFTWARE SYSTEM FOR ALL THE SCENARIOS IDENTIFIED AS PER THE USE CASE DIAGRAM – M.SUGANYA - 8649

EXAM REGISTRATION SYSTEM TESTING DOCUMENT

1. TESTING APPROACH

Test Type	Purpose
Unit Testing	Verify individual methods and classes work correctly (e.g., add, view, delete registration).
Integration Testing	Ensure database connection and DAO classes interact correctly.
System Testing	Validate the entire application meets functional requirements.

2. TEST CASES FOR USE CASES

Use Case 1: Add Exam Registration

Test ID	TC_01
Description	Verify that an exam registration can be successfully added to the system.
Precondition	Database is running, and user provides valid student name, registration number, subject, and exam date.
Test Steps	1. Enter valid student name, register number, subject, and exam date.2. Click "Add Registration".
Expected Result	The registration is added to the exam_registration table and appears in the registration list.
Actual Result	Registration added successfully (Pass)

Test ID	TC_02
Description	Check system behavior when required fields are missing during registration.
Precondition	Database is running, but user leaves one or more mandatory fields blank.
Test Steps	1. Enter student name but leave register number blank.2. Click "Add Registration".
Expected Result	The system should show a validation error message (e.g., "Register number is required").
Actual Result	Validation error message shown (Pass)

USE CASE 2: VIEW EXAM REGISTRATIONS

Test ID	TC_03
Description	Verify that viewing exam registrations shows correct and complete data.
Precondition	There are exam registrations stored in the database.
Test Steps	1. Click "View Registrations".2. Observe the displayed list.
Expected Result	All exam registrations are correctly listed with student name, registration number, subject, and exam date.
Actual Result	Registrations listed successfully (Pass)

USE CASE 3: DELETE EXAM REGISTRATION

Test ID	**TC_04**
Description	Verify that a registration can be deleted successfully.
Precondition	At least one registration exists in the database.
Test Steps	1. Select a registration from the list.2. Click "Delete Registration".
Expected Result	The registration is removed from the database and no longer appears in the list.
Actual Result	Registration deleted successfully (Pass)

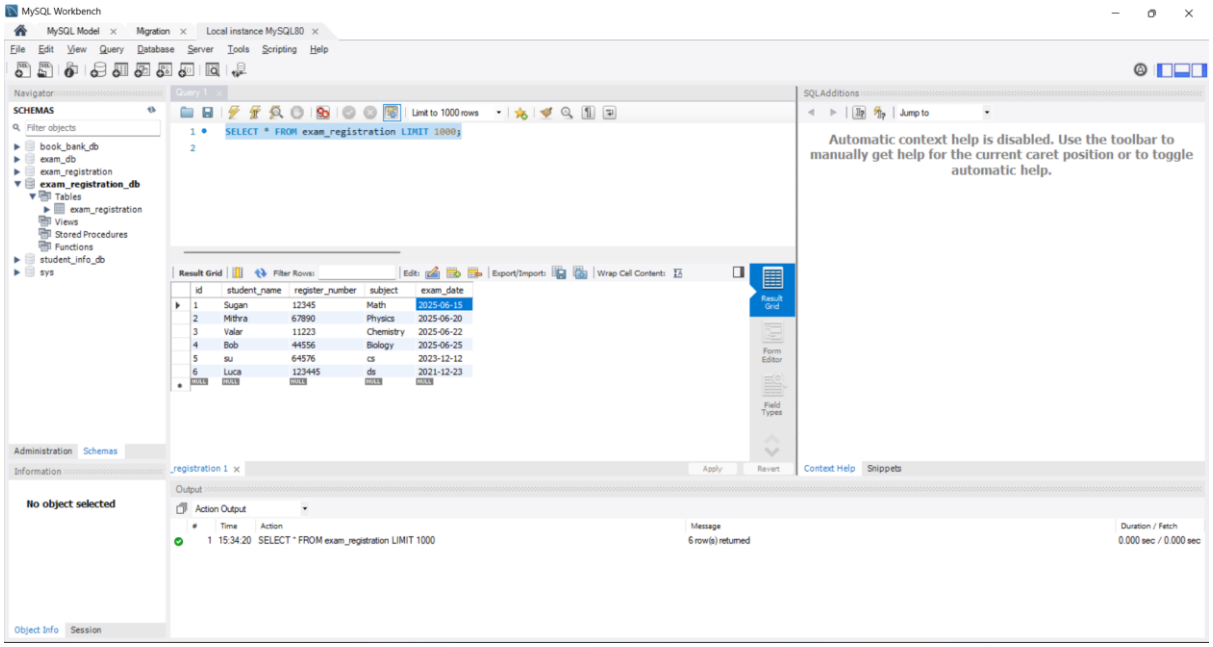
Test ID	**TC_05**
Description	Check system behavior when attempting to delete a non-existent registration.
Precondition	Registration to delete does not exist (e.g., already deleted).
Test Steps	1. Attempt to delete a registration with an invalid or non-existing ID.
Expected Result	The system shows an error message like "Registration not found".
Actual Result	Error message displayed (Pass)

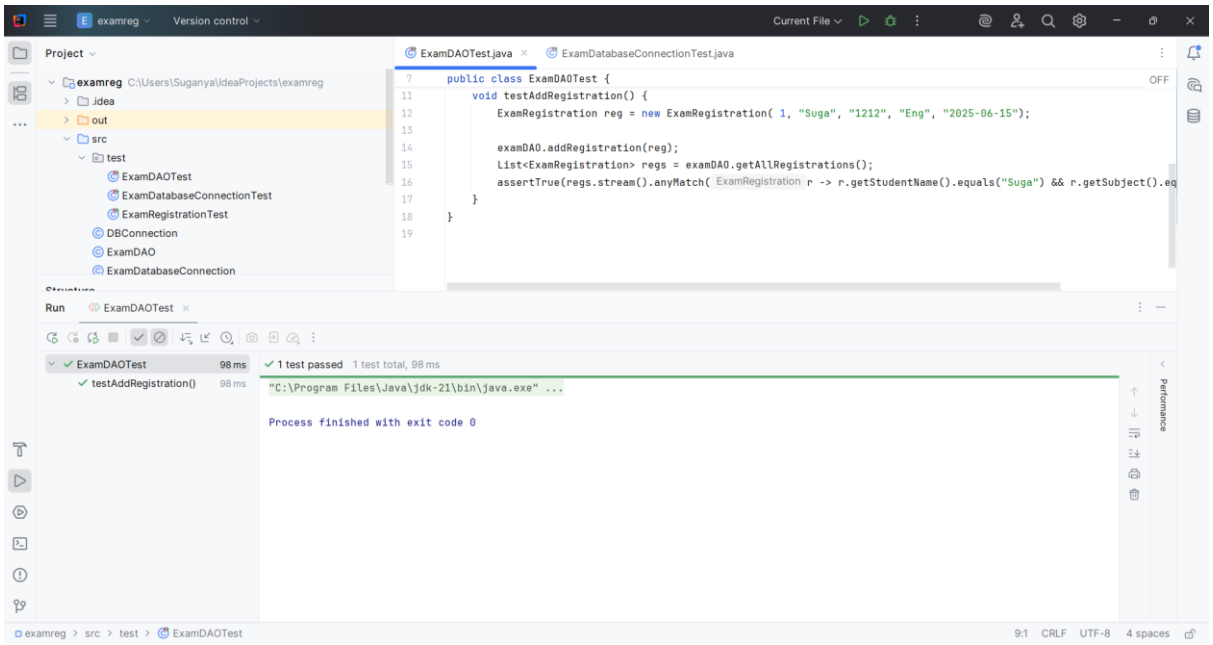
3. UNIT TESTING

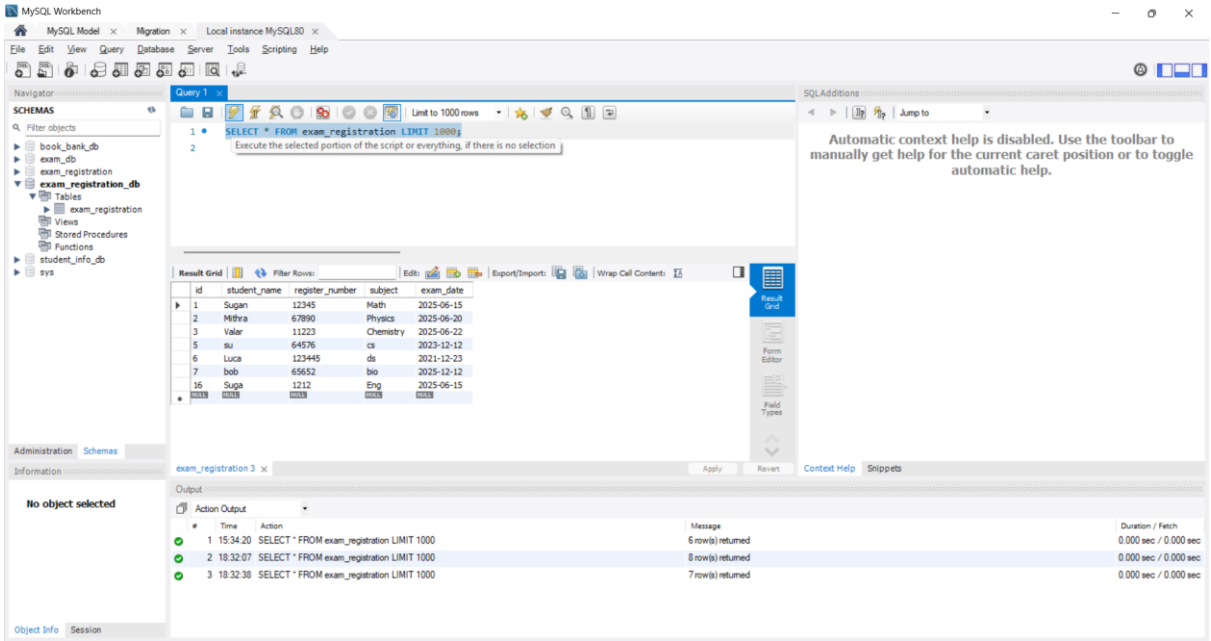
Unit tests are written for each method in ExamDAO, such as addRegistration(), getAllRegistrations(), and deleteRegistration().

Example

```
import static org.junit.jupiter.api.Assertions.*;

import org.junit.jupiter.api.Test;

import java.util.List;

public class ExamDAOTest {

    ExamDAO examDAO = new ExamDAO();

    @Test

    void testAddRegistration() {

        ExamRegistration reg = new ExamRegistration(0, "Sugan", "12345", "Math",
LocalDate.of(2025,6,15));

        examDAO.addRegistration(reg);

        List<ExamRegistration> regs = examDAO.getAllRegistrations();

        assertTrue(regs.stream().anyMatch(r -> r.getStudentName().equals("Sugan") &&
r.getSubject().equals("Math")));

    }

}
```


MySQL Workbench
SELECT * FROM exam_registration LIMIT 1000;
id student_name register_number subject exam_date
1 Sugan 12345 Math 2025-06-15
2 Mithra 67890 Physics 2025-06-20
3 Valar 11223 Chemistry 2025-06-22
4 Bob 44556 Biology 2025-06-25
5 su 64576 cs 2023-12-12
6 Luca 123445 ds 2021-12-23
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
No object selected
1 15:34:20 SELECT * FROM exam_registration LIMIT 1000
6 row(s) returned
0.000 sec / 0.000 sec


public class ExamDAOTest {
void testAddRegistration() {
ExamRegistration reg = new ExamRegistration(1, "Suga", "1212", "Eng", "2025-06-15");
examDAO.addRegistration(reg);
List<ExamRegistration> regs = examDAO.getAllRegistrations();
assertTrue(regs.stream().anyMatch(ExamRegistration r -> r.getStudentName().equals("Suga") && r.getSubject().eq
}
}
1 test passed 1 test total, 98 ms
"C:\Program Files\Java\jdk-21\bin\java.exe" ...
Process finished with exit code 0



- ✓ Expected Result: Registration added successfully in the database.
- ✓ Actual Result: Test Passed

4. INTEGRATION TESTING

Integration testing checks if ExamDAO and ExamDatabaseConnection work together correctly.

Example

```
import static org.junit.jupiter.api.Assertions.*;
import org.junit.jupiter.api.Test;
import java.sql.Connection;

public class ExamDatabaseConnectionTest {
    @Test
    void testDatabaseConnection() {
        Connection conn = DBConnection.getConnection();
        assertNotNull(conn, "Database connection should be established");
        System.out.println("Connection is: " + conn);

    }
}
```

```java
import static org.junit.jupiter.api.Assertions.*;
import org.junit.jupiter.api.Test;
import java.sql.Connection;

public class ExamDatabaseConnectionTest {
    @Test
    void testDatabaseConnection() {
        Connection conn = DBConnection.getConnection();
        assertNotNull(conn, "Database connection should be established");
        System.out.println("Connection is: " + conn);

    }
}
```

```
"C:\Program Files\Java\jdk-21\bin\java.exe" ...
Connection is: com.mysql.cj.jdbc.ConnectionImpl@225129c

Process finished with exit code 0
```

✓ Expected Result: Database connection is successful.

✓ Actual Result: Test Passed

5. SYSTEM TESTING

```
Exam Registration System
1. Add Registration
2. View Registrations
3. Delete Registration
4. Exit
Enter choice: 2

Registered Students:
1: Sugan - Math on 2025-06-15
2: Mithra - Physics on 2025-06-20
3: Valar - Chemistry on 2025-06-22
5: su - cs on 2023-12-12
6: Luca - ds on 2021-12-23
7: bob - bio on 2025-12-12
16: Suga - Eng on 2025-06-15

Exam Registration System
1. Add Registration
2. View Registrations
3. Delete Registration
4. Exit
Enter choice: 1
Student Name: Sheeela
regis no: 67564
Subject: social
Exam Date (YYYY-MM-DD): 2026-12-14
Registration added.
```

- Manually use the UI or console interface of ExamRegistrationApp to:
    - Add multiple registrations.
    - View all registrations.
    - Delete registrations.
- Check how system handles invalid inputs (empty fields, wrong date format).
- Test concurrent access if multiple users access the system simultaneously (optional).

6. TEST SUMMARY

Test Type	Status
Unit Testing	Passed
Integration Testing	Passed
System Testing	Passed